

AD 502 – Machine Learning

Pre-Reading Material – Lecture 13

Relation Between AI, ML, DL and DS

Estimated Reading Time: 25–30 minutes

Why This Topic Matters

Artificial Intelligence, Machine Learning, Deep Learning, and Data Science are closely related fields, but they are **not the same thing**.

These terms are often used interchangeably, which can create confusion. Understanding their relationship helps you see where **Machine Learning fits within AI**, how **Deep Learning fits within Machine Learning**, and how **Data Science uses many of these techniques to extract useful insights from data**.

Learning Outcomes

After completing this pre-reading, you should be able to:

- Explain the basic concept of **Artificial Intelligence (AI)**.
- Explain the role of **Machine Learning (ML)** within AI.
- Understand how **Deep Learning (DL)** is related to ML.
- Explain the purpose of **Data Science (DS)**.
- Differentiate between AI, ML, DL, and DS.
- Identify how these fields overlap in real-world applications.

Activate Your Prior Knowledge

1. Artificial Intelligence

You have already learned that AI aims to develop systems capable of performing tasks that normally require human intelligence.

Think:



How can a computer learn to recognize a face without being explicitly programmed with rules for every possible face?

2. Machine Learning

You have studied that ML allows systems to learn patterns from data.

Think:

Is every AI system necessarily a Machine Learning system?

3. Data

Machine Learning depends heavily on data.

Think:

If we have a large dataset, can simply analyzing the data be called Machine Learning?

Prerequisite Concepts

You should be familiar with:

- Artificial Intelligence
- Machine Learning
- Data and datasets
- Supervised and unsupervised learning
- Basic data analysis

Core Concepts

1. What is Artificial Intelligence (AI)?

Artificial Intelligence (AI) is the broader field concerned with creating systems that can perform tasks requiring capabilities associated with human intelligence.

Examples include:

- Reasoning
- Decision-making



- Problem-solving
- Understanding language
- Perception
- Learning

AI can use different approaches, including **rule-based systems, search, optimization, Machine Learning, and Deep Learning.**

Therefore:

AI is the broader field.

2. What is Machine Learning (ML)?

Machine Learning is a subfield of AI in which systems learn patterns from data and use those patterns to make predictions or decisions.

Instead of explicitly programming every rule, we provide:

Data → Learning Algorithm → Model → Prediction/Decision

Examples include:

- Spam detection
- House-price prediction
- Recommendation systems
- Fraud detection

Therefore:

Machine Learning is one approach used to build AI systems.

3. What is Deep Learning (DL)?

Deep Learning is a specialized area of Machine Learning that uses **multi-layer neural networks** to learn complex patterns from data.

A simplified relationship is:

Artificial Intelligence



Machine Learning



Deep Learning

Deep Learning has been particularly successful in areas involving large and complex datasets such as:

- Image recognition
- Speech recognition
- Natural language processing
- Generative AI

Therefore:

Deep Learning is a subset of Machine Learning.

4. What is Data Science (DS)?

Data Science is an interdisciplinary field concerned with obtaining useful information and insights from data.

It can involve:

- Data collection
- Data cleaning
- Data exploration
- Statistical analysis
- Visualization
- Machine Learning
- Communication of results

A simplified Data Science workflow is:

Collect Data



Clean & Prepare Data



Explore & Analyze



Build Models



Interpret Results



Communicate Insights

Therefore:

Data Science is broader than simply building Machine Learning models.

5. Relationship Between AI, ML and DL

The relationship can be visualized as:

Artificial Intelligence

└─ **Machine Learning**

└─ **Deep Learning**

This means:

- AI is the broader concept.
- ML is a subset of AI.
- DL is a subset of ML.

However, **not all AI uses ML**, and **not all ML uses Deep Learning**.



For example, a rule-based expert system can be considered an AI system without using Machine Learning.

6. Relationship Between Data Science and ML

Data Science and Machine Learning overlap significantly.

A Data Scientist may use Machine Learning to:

- Predict future outcomes
- Classify observations
- Discover patterns
- Segment customers
- Detect anomalies

However, Data Science also includes activities that are not necessarily Machine Learning, such as:

- Data cleaning
- Exploratory data analysis
- Visualization
- Statistical analysis
- Reporting and communication

Therefore:

ML can be a tool within Data Science, but Data Science is not limited to ML.

7. AI, ML, DL and DS – Simple Comparison

Field Main Focus

Example

AI	Making machines intelligent	Intelligent assistant
ML	Learning patterns from data	Spam detection
DL	Learning complex patterns using deep neural networks	Image recognition
DS	Extracting insights and value from data	Customer analysis

8. A Real-World Example

Consider a **self-driving car**.

AI

The overall goal is to make the vehicle behave intelligently.

ML

Machine Learning can learn patterns for tasks such as predicting road conditions or other vehicles' behavior.

DL

Deep Learning can process complex inputs such as camera images and recognize:

- Vehicles
- Pedestrians
- Traffic signs
- Road markings

Data Science

Data Science can be used to:

- Analyze driving data
- Clean datasets
- Identify patterns
- Evaluate model performance
- Generate insights

Thus, different fields can work together in the same application.

Guided Reading Questions

1. What is **Artificial Intelligence**?
2. Is every AI system necessarily based on Machine Learning?
3. What is **Machine Learning**?
4. What is **Deep Learning**?
5. Why is Deep Learning considered a subset of Machine Learning?

6. What is **Data Science**?
7. How does Data Science use Machine Learning?
8. Is Data Science limited to Machine Learning?
9. What is the relationship between **AI, ML, and DL**?
10. Give one real-world example where AI, ML, DL, and DS can work together.

Reflection Activity

Write 75–100 words explaining the relationship between **AI, ML, DL, and Data Science**. Use one real-world application to show how these fields can overlap.

Self-Assessment

1. The broader field concerned with creating intelligent machines is called _____.
2. Machine Learning is a subset of _____.
3. Deep Learning is a subset of _____.
4. Deep Learning primarily uses multi-layer _____ networks.
5. _____ Science focuses on extracting useful insights from data.
6. Spam detection is an example of a common _____ application.
7. Image recognition is an important application of _____ Learning.
8. Data cleaning and visualization are activities commonly associated with _____ Science.
9. Every AI system necessarily uses Machine Learning. **True / False**
10. Every Deep Learning system is a Machine Learning system. **True / False**

Key Takeaway

The most important relationship to remember is:

AI



ML



DL

- **AI** is the broader field of creating intelligent systems.
- **ML** enables systems to learn patterns from data.



- **DL** is a specialized form of ML based on deep neural networks.
- **DS** focuses on extracting useful information and insights from data and may use ML and DL as tools.

AI is the broader goal, ML is one way to achieve AI, DL is a specialized approach within ML, and Data Science uses data, statistics, programming, and often ML to generate insights and solutions.

